Event-Aligned Singular Value Decomposition for Data-Driven Pattern Kernel Discovery in Cryptocurrency Perpetual Futures

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Abstract

We propose a data-driven method for discovering predictive multi-candle price patterns in high-frequency cryptocurrency markets. The pipeline defines six analytical event types (breakouts, turtle-soup false breakouts, bull/bear traps), extracts OHLCV windows aligned to each event, decomposes each window into four signal channels (body, upper wick, lower wick, volume), and applies Singular Value Decomposition to discover the dominant shape variations. A subsequent Information Coefficient gate with Benjamini–Hochberg false discovery rate control identifies which discovered shapes correlate with forward returns. The method inherits the interpretability of classical technical analysis (events are analytically defined) while leveraging the statistical efficiency of data-driven basis discovery (shapes are learned, not hand-designed). Crucially, SVD operates on price geometry alone without access to forward returns, providing a structural safeguard against overfitting. Applying the pipeline to BTC perpetual futures tick data (7.2 million ticks, 12,059 one-minute candles), we discover 6 statistically significant pattern kernels from 158 candidates after FDR correction ($q = 0.05$). Turtle-soup and trap event types produce the strongest signals ($|\text{IC}| \in [0.20, 0.36]$), while breakout patterns show no predictive power after multiple testing correction. Rolling SVD stability analysis confirms that the discovered shapes are temporally persistent ($\bar{\rho} > 0.85$ for the strongest kernels), supporting their use as real-time features via cosine similarity scoring.

Keywords: pattern recognition, singular value decomposition, market microstructure, technical analysis, cryptocurrency, matched filtering, event-aligned averaging

JEL Classification: C14, C58, G14, G17

Contents

1 Introduction

1.1 The Problem

Classical technical analysis assigns fixed geometric shapes to named patterns: a "head and shoulders" has a specific form; a "hammer" has a long lower wick and a small body. These hand-designed templates carry two fundamental limitations: (1) the actual shape of a pattern in a given market and regime may differ from the textbook version, and (2) there is no principled mechanism for deciding *which* shape variation predicts future returns as opposed to merely being visually recognisable.

This paper addresses both limitations. We define market events *analytically* by price action conditions (Section 5), making event detection unambiguous and replicable. The shapes associated with those events are then *discovered from data* via Singular Value Decomposition (Section 7). Only the discovered shapes that survive a statistical Information Coefficient gate (Section 8) are retained. The result is a kernel library that is both data-driven and statistically validated.

1.2 Analogues in Other Disciplines

The methodology draws on two established signal processing paradigms.

Event-Related Potentials (ERP) in neuroscience. An ERP is computed by locking EEG recordings to a stimulus event and averaging across trials. The average converges to the signal phase-locked to the event, while trial-to-trial noise cancels. Our event-aligned matrix $\mathbf{X}^{(e,c)}$ plays exactly this role: each row is one "trial" (one occurrence of the event), and SVD generalises simple averaging by retaining not just the mean shape but the full low-dimensional subspace of shape variation. The left singular vector $\mathbf{u}_1$ encodes trial-to-trial amplitude variation; the right singular vector $\mathbf{v}_1$ encodes the dominant waveform shape. This connection to neuroscience ERP methodology is, to our knowledge, novel in the financial pattern recognition literature.

Matched filtering in radar and seismology. A matched filter maximises the signal-to-noise ratio for detecting a known template $\mathbf{k}$ in a noisy signal $\mathbf{x}(t)$ by computing the cross-correlation $s(t) = \mathbf{k}^\top \mathbf{x}(t)$ (Turin, 1960). The SVD right singular vectors $\mathbf{v}_k$ serve as data-driven templates, and the online scoring step (Section 9) is exactly a matched filter bank—one filter per surviving component.

1.3 Key Insight: Shape Discovery Before Return Fitting

The critical anti-overfitting property is the ordering of operations:

1. SVD discovers basis functions using only the *event-aligned price data.* It has no access to forward returns.

2. IC filtering checks which of these already-discovered shapes correlate with forward returns.

We are not fitting shapes to returns. We are asking whether naturally occurring shape variation—variation the market itself exhibits around analytically-defined events—is predictive. This ordering ensures that the discovered kernels are interpretable geometric objects, not overfit regression artefacts.

1.4 Contributions

1. A complete pipeline for data-driven pattern kernel discovery that combines analytical event definitions, multi-channel OHLCV decomposition, SVD basis discovery, and IC-based statistical filtering with FDR control.

2. A multi-channel candlestick representation (body, upper wick, lower wick, volume) that enables SVD to capture shape variation independently across price geometry and volume dynamics.

3. Empirical evidence from BTC perpetual futures showing that turtle-soup and trap event types produce temporally stable, statistically significant pattern kernels, while breakout patterns do not survive multiple testing correction.

4. An analytical basis dictionary (10 functions) for interpretability diagnostics and warm-start initialisation, together with evidence that data-driven shapes are genuinely novel (maximum alignment < 0.5 with any analytical basis).

2 Related Work

Technical pattern recognition. Lo, Mamaysky, and Wang (2000) provided the first rigorous statistical evaluation of technical analysis, using kernel regression to detect classical patterns (head-and-shoulders, triangles, double tops) and testing their information content via bootstrap. Their approach identifies predefined pattern types; ours discovers shapes endogenously via SVD. Leigh *et al.* (2002) used template matching with fixed bull-flag templates. Chang and Osler (1999) studied the profitability of head-and-shoulders patterns in foreign exchange. All of these methods rely on hand-designed templates, which our approach replaces with data-driven discovery.

PCA and random matrix theory in finance. The application of eigenvalue decomposition to financial correlation matrices is well established (Laloux *et al.*, 1999; Plerou *et al.*, 2002; Bouchaud and Potters, 2003). These works decompose cross-sectional return covariance to separate signal from noise. Our application differs fundamentally: we apply SVD to *time-domain* event-aligned windows of a single asset's candle data, decomposing temporal shape variation rather than cross-sectional correlation structure.

Event studies. The event study methodology (Fama *et al.*, 1969; MacKinlay, 1997) examines abnormal returns around corporate events (earnings, mergers). The standard approach averages returns across events; it does not examine the *shape* of price action preceding the event. Our event-aligned matrix construction is analogous to cumulative abnormal return (CAR) windows but applied to raw OHLCV channels and decomposed via SVD rather than averaged.

Signal processing in finance. Cont (2001) introduced the concept of empirical scaling laws and statistical properties of financial time series. Gatheral and Oomen (2010) connected microstructure noise to realised volatility estimation. The matched filtering analogy we draw on originates in radar signal detection (Turin, 1960) and gravitational wave astronomy (Allen *et al.*, 2012), where template banks of known waveforms are correlated against noisy detector output. Our contribution adapts this framework by constructing the template bank from data via SVD rather than from physical models.

Cryptocurrency microstructure. Makarov and Schoar (2020) studied price formation and arbitrage in cryptocurrency markets. Ait-Sahalia and Brunetti (2024) analysed order flow toxicity in Bitcoin markets. Our work contributes to this literature by providing a systematic method for extracting predictive candle-level patterns from high-frequency cryptocurrency data.

3 Notation

Symbol	Type / Range	Meaning
t	$t \in \mathbb{N}$	Discrete candle time index
O_t, H_t, L_t, C_t	$\mathbb{R}_{>0}$	Open, High, Low, Close prices at bar t
V_t	$\mathbb{R}_{\geq 0}$	Traded volume at bar t
b_t, u_t, l_t, v_t	$\mathbb{R}$	Body, upper wick, lower wick, volume channels
W	$W \in \mathbb{N}$	Window width in candles
N	$N \in \mathbb{N}$	Lookback for event detection
K	$K \in \mathbb{N}$	Confirmation lag (trap events)
α	$\alpha > 0$	ATR multiplier for trap threshold
n_e	$n_e \in \mathbb{N}$	Number of events of type e
$\mathcal{E}$	set	Set of event types (6 elements)
$\mathcal{C}$	$\{b, u, l, v\}$	Set of signal channels
τ	$\{0, \ldots, W-1\}$	Within-window time index
$\bar{\tau}$	$[0, 1]$	Normalised window coordinate
$\mathbf{X}^{(e,c)}$	$\mathbb{R}^{n_e \times W}$	Event-aligned data matrix
$\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}$	—	SVD factors
σ_k	$\mathbb{R}_{\geq 0}$	k-th singular value
$\mathbf{v}_k$	$\mathbb{R}^W$	k-th right singular vector (basis shape)
$\mathbf{u}_k$	$\mathbb{R}^{n_e}$	k-th left singular vector (event loadings)
EVR_k	$[0, 1]$	Explained variance ratio of component k
SI	$\mathbb{R}_{\geq 0}$	Stereotypy index σ_1/σ_2
IC_k	$[-1, 1]$	Information Coefficient for component k
$r_i^{(H)}$	$\mathbb{R}$	Forward log-return at horizon H
$s_j^{(e,c)}(t)$	$[-1, 1]$	Cosine-similarity score
$S^{(e)}(t)$	$\mathbb{R}$	Aggregated score for event type e
w_c	$w_c \geq 0$	Channel aggregation weight
ATR_t	$\mathbb{R}_{>0}$	Average True Range at time t
ϕ_i	$[0, 1] \to \mathbb{R}$	Analytical basis function i

4 Candlestick Decomposition into Signal Channels

4.1 OHLCV Decomposition

A single OHLCV candle at time t is the tuple $(O_t, H_t, L_t, C_t, V_t)$ satisfying

$$L_t \leq \min(O_t, C_t) \leq \max(O_t, C_t) \leq H_t, \quad V_t \geq 0. \tag{1}$$

We decompose each candle into four *signal channels*.

Definition 4.1 (Signal Channels)**.** For a candle at time t, define:

$$b_t := C_t - O_t \tag{2}$$

$$u_t := H_t - \max(C_t,\, O_t) \tag{3}$$

$$l_t := \min(C_t,\, O_t) - L_t \tag{4}$$

$$v_t := V_t \tag{5}$$

Remark 4.1. By (1), $u_t \geq 0$ and $l_t \geq 0$ always. The body b_t is signed: $b_t > 0$ for a bull candle, $b_t < 0$ for a bear candle. The decomposition is invertible: given (b_t, u_t, l_t, O_t) one recovers (O_t, H_t, L_t, C_t) exactly. The channel set is $\mathcal{C} = \{b, u, l, v\}$.

The multi-channel representation is essential: different event types exhibit their characteristic signatures in different channels. As we show empirically (Section 12), the volume channel carries independent predictive information for trap events that is absent from the price channels alone.

4.2 Window Extraction

Definition 4.2 (Event Window)**.** For an event detected at time t_e and window width $W \in \mathbb{N}$, the *event window* of channel $c \in \mathcal{C}$ is the vector

$$\mathbf{x}^{(c)}(t_e) := \left(x^{(c)}_{t_e-W+1},\ x^{(c)}_{t_e-W+2},\ \ldots,\ x^{(c)}_{t_e}\right)^\top \in \mathbb{R}^W, \tag{6}$$

where $x^{(c)}_t \in \{b_t, u_t, l_t, v_t\}$ for $c \in \{b, u, l, v\}$ respectively.

4.3 Per-Window Normalisation

Raw price levels are not comparable across events separated in time or across volatility regimes. We apply a *shape-preserving* normalisation that removes level and scale but retains geometry.

Definition 4.3 (ATR Normalisation)**.** Let ATR_t be the N_{atr}-period Average True Range (Wilder, 1978):

$$\mathrm{ATR}_t := \frac{1}{N_{\mathrm{atr}}} \sum_{j=0}^{N_{\mathrm{atr}}-1} \max\bigl(H_{t-j} - L_{t-j},\ |H_{t-j} - C_{t-j-1}|,\ |L_{t-j} - C_{t-j-1}|\bigr). \tag{7}$$

The normalised window for event i with event time $t_e^{(i)}$ is

$$\tilde{x}^{(c)}_i(\tau) := \frac{x^{(c)}_{t_e^{(i)}-W+1+\tau} - \mu^{(c)}_i}{\mathrm{ATR}_{t_e^{(i)}}}, \quad \tau = 0, 1, \ldots, W-1, \tag{8}$$

where $\mu^{(c)}_i := \frac{1}{W} \sum_{\tau=0}^{W-1} x^{(c)}_{t_e^{(i)}-W+1+\tau}$ is the within-window mean.

5 Analytical Event Definitions

We define six event types in $\mathcal{E}$. Let N be the lookback period and let ATR_t be as in (7).

5.1 Breakout Events

Definition 5.1 (Breakout, Bull)**.** A *bull breakout* occurs at time t if and only if

$$C_t > \max_{j=1}^{N} H_{t-j} \quad \text{and} \quad V_t > 1.5 \cdot \mathrm{median}\bigl(V_{t-N}, \ldots, V_{t-1}\bigr). \tag{9}$$

Definition 5.2 (Breakout, Bear)**.** A *bear breakout* occurs at time t if and only if

$$C_t < \min_{j=1}^{N} L_{t-j} \quad \text{and} \quad V_t > 1.5 \cdot \mathrm{median}\bigl(V_{t-N}, \ldots, V_{t-1}\bigr). \tag{10}$$

The volume condition filters genuine momentum moves from slow drift: a real breakout typically registers at least $1.5\times$ the recent median volume. The N-period range maximum $\max_{j=1}^{N} H_{t-j}$ is the upper Donchian channel (Donchian, 1960), excluding the current bar.

5.2 Turtle Soup Events (False Range Break)

Turtle Soup refers to a pattern described by Raschke and Connors (1995): price transiently penetrates an N-period extreme, then closes back inside the prior range, suggesting that the breakout attracted stop-loss orders absorbed by a larger counterparty.

Definition 5.3 (Turtle Soup, Bull)**.** A *bull turtle soup* (fake breakdown, anticipates reversal long) occurs at time t if and only if

$$L_t < \min_{j=1}^{N} L_{t-j} \quad \text{and} \quad C_t > \min_{j=1}^{N} L_{t-j}. \tag{11}$$

Definition 5.4 (Turtle Soup, Bear)**.** A *bear turtle soup* (fake breakout, anticipates reversal short) occurs at time t if and only if

$$H_t > \max_{j=1}^{N} H_{t-j} \quad \text{and} \quad C_t < \max_{j=1}^{N} H_{t-j}. \tag{12}$$

In both cases the candle's extreme pierces the historical range but the close reverts. The lower wick channel l_t is expected to be especially large for Definition 5.3, providing a signal-rich feature for SVD.

5.3 Trap Events (Confirmed False Breakout)

A trap event requires multi-bar confirmation: the breakout occurs at time t, and the close at time $t+K$ must have reversed by at least $\alpha \cdot \mathrm{ATR}_t$.

Definition 5.5 (Bull Trap)**.** A *bull trap* is a pair $(t, t+K)$ satisfying:

(i) Condition (9) holds at time t.

(ii) $C_{t+K} < C_t - \alpha \cdot \mathrm{ATR}_t$.

The event window is centred on the breakout candle at time t.

Definition 5.6 (Bear Trap)**.** A *bear trap* is a pair $(t, t+K)$ satisfying:

(i) Condition (10) holds at time t.

(ii) $C_{t+K} > C_t + \alpha \cdot \mathrm{ATR}_t$.

Remark 5.1 (Look-ahead bias)**.** Trap events use information from $t+K$ to label the event at t. They are therefore only valid for offline kernel discovery. The aligned window is extracted at t, and the confirmation at $t+K$ partitions events into "trap" versus "genuine breakout" classes. Deploying the kernel in real time introduces no look-ahead, because the kernel scores the current window at bar t without knowledge of future prices.

Parameter defaults.

Parameter	**Symbol**	**Default**	**Valid Range**
Range lookback	N	20 bars	$[10, 60]$
Volume multiplier	—	1.5	$[1.2, 2.5]$
ATR period	N_{atr}	14 bars	$[7, 28]$
Trap confirmation lag	K	3 bars	$[1, 10]$
ATR reversal threshold	α	1.0	$[0.5, 2.0]$

6 Event-Aligned Matrix Construction

Definition 6.1 (Event-Aligned Matrix)**.** Let $\mathcal{T}_e = \{t_e^{(1)}, t_e^{(2)}, \ldots, t_e^{(n_e)}\}$ be the ordered set of detection times for event type $e \in \mathcal{E}$. For channel $c \in \mathcal{C}$, the *event-aligned matrix* is

$$\mathbf{X}^{(e,c)} := \begin{pmatrix} \tilde{x}_1^{(c)}(0) & \tilde{x}_1^{(c)}(1) & \cdots & \tilde{x}_1^{(c)}(W-1) \\ \tilde{x}_2^{(c)}(0) & \tilde{x}_2^{(c)}(1) & \cdots & \tilde{x}_2^{(c)}(W-1) \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{x}_{n_e}^{(c)}(0) & \tilde{x}_{n_e}^{(c)}(1) & \cdots & \tilde{x}_{n_e}^{(c)}(W-1) \end{pmatrix} \in \mathbb{R}^{n_e \times W}, \tag{13}$$

where $\tilde{x}_i^{(c)}(\tau)$ is given by (8).

Each row is one normalised event window. Each column corresponds to a relative time offset τ within the window. We require

$$n_e \geq n_{\min} = 100 \tag{14}$$

before proceeding to decomposition. Under a spiked covariance model, the variance of the estimated right singular vector is $O(\sigma^2 W/(n_e \sigma_k^2))$; for $W = 20$ and typical signal-to-noise ratios, $n_e = 100$ achieves $O(10\%)$ estimation error.

7 SVD Decomposition: Basis Discovery

7.1 The Decomposition

Definition 7.1 (Thin SVD)**.** For $\mathbf{X} := \mathbf{X}^{(e,c)} \in \mathbb{R}^{n_e \times W}$ with $n_e \geq W$, the thin Singular Value Decomposition is

$$\mathbf{X} = \mathbf{U}\,\mathbf{\Sigma}\,\mathbf{V}^\top, \tag{15}$$

where $\mathbf{U} \in \mathbb{R}^{n_e \times W}$ with $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_W$, $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \ldots, \sigma_W)$ with $\sigma_1 \geq \cdots \geq \sigma_W \geq 0$, and $\mathbf{V} \in \mathbb{R}^{W \times W}$ with $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_W$.

The columns $\mathbf{v}_k = \mathbf{V}_{:,k} \in \mathbb{R}^W$ are the *discovered basis functions* (shapes), ordered from dominant to residual. The columns $\mathbf{u}_k = \mathbf{U}_{:,k} \in \mathbb{R}^{n_e}$ give the *per-event loading* on shape k: the scalar u_{ik} measures how strongly event i projects onto shape k.

7.2 Interpretive Quantities

Definition 7.2 (Explained Variance Ratio)**.**

$$\mathrm{EVR}_k := \frac{\sigma_k^2}{\sum_{j=1}^{W} \sigma_j^2} = \frac{\sigma_k^2}{\|\mathbf{X}\|_F^2}. \tag{16}$$

Definition 7.3 (Stereotypy Index)**.** The *stereotypy index* of event class (e, c) is

$$\mathrm{SI}^{(e,c)} := \frac{\sigma_1}{\sigma_2}. \tag{17}$$

A high stereotypy index ($\mathrm{SI} \gg 1$) indicates a single dominant shape; an index near 1 indicates a broad, multi-modal shape distribution.

7.3 Truncation Rule

We retain the top $K^{(e,c)}$ components where cumulative explained variance first exceeds $\rho^* = 0.80$:

$$K^{(e,c)} := \min\left\{ K \in \{1, \ldots, W\} \;:\; \sum_{k=1}^{K} \mathrm{EVR}_k \geq \rho^* \right\}. \tag{18}$$

By the Eckart–Young–Mirsky theorem (Eckart and Young, 1936), the retained components provide the best rank-K approximation to $\mathbf{X}$ in Frobenius norm, with relative reconstruction error $\epsilon_K = \sqrt{1 - \sum_{k=1}^{K} \mathrm{EVR}_k}$.

8 Predictive Filtering: The IC Gate

8.1 Forward Returns

Definition 8.1 (Forward Log-Return)**.** For event i at time $t_e^{(i)}$ and horizon H candles:

$$r_i^{(H)} := \ln \frac{C_{t_e^{(i)}+H}}{C_{t_e^{(i)}}}. \tag{19}$$

The return begins strictly after the event window ends, ensuring no overlap between the information used by SVD and the return being predicted.

8.2 Information Coefficient

Definition 8.2 (Information Coefficient, IC)**.** For SVD component k of event type (e, c):

$$\mathrm{IC}_k^{(e,c,H)} := \mathrm{corr}\big(\mathbf{u}_k,\ \mathbf{r}^{(H)}\big). \tag{20}$$

Component k is retained if $|\mathrm{IC}_k^{(e,c,H)}| \geq \delta_{\mathrm{IC}} = 0.03$.

Remark 8.1 (Why this is not overfitting)**.** The loadings $\mathbf{u}_k$ are determined entirely by the price-action matrix $\mathbf{X}^{(e,c)}$. The SVD objective minimises reconstruction error with no knowledge of $\mathbf{r}^{(H)}$. The IC computation is a *test* on the SVD output, not a training step. This is analogous to computing the correlation between a PCA projection and an external label.

8.3 Multiple Comparison Correction

Across all combinations (e, c, k, H), the IC tests constitute a family of hypotheses. We apply Benjamini–Hochberg FDR control (Benjamini and Hochberg, 1995) at level $q = 0.05$:

1. Compute the two-sided p-value under H_0: $\mathrm{IC} = 0$ using the t-statistic

$$t_k = \mathrm{IC}_k \sqrt{\frac{n_e - 2}{1 - \mathrm{IC}_k^2}}, \quad t_k \sim t_{n_e - 2} \text{ under } H_0. \tag{21}$$

2. Order p-values: $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$.
3. Reject H_0 for all $j \leq j^*$, where $j^* := \max\{j : p_{(j)} \leq jq/m\}$.

9 Kernel Construction and Online Scoring

9.1 Kernel Library

After IC filtering, the kernel library is

$$\mathcal{K} := \big\{\mathbf{k}_j^{(e,c)} \in \mathbb{R}^W \ :\ (e, c, j) \text{ survives IC gate and BH-FDR}\big\}, \tag{22}$$

where $\mathbf{k}_j^{(e,c)} := \mathbf{v}_j^{(e,c)}$. Each kernel is unit norm, orthogonal to other kernels within the same (e, c) class, and mean-free by construction.

9.2 Cosine Similarity Scoring

At production time t, the current normalised window for channel c is $\mathbf{x}_t^{(c)} \in \mathbb{R}^W$. The cosine similarity score is

$$s_j^{(e,c)}(t) := \frac{\langle \mathbf{x}_t^{(c)}, \mathbf{k}_j^{(e,c)} \rangle}{\left\| \mathbf{x}_t^{(c)} \right\|_2}, \tag{23}$$

where the simplification uses $\|\mathbf{k}\|_2 = 1$. The score satisfies $s \in [-1, 1]$ by the Cauchy–Schwarz inequality.

9.3 Multi-Channel Aggregation

For event type e, the composite score aggregates across channels and surviving components:

$$S^{(e)}(t) := \sum_c w_c \sum_{\{j:\mathbf{k}_j^{(e,c)}\in\mathcal{K}\}} \mathrm{IC}_j \cdot s_j^{(e,c)}(t), \tag{24}$$

where w_c are channel weights ($w_c = 0.25$ by default or $w_c \propto |\mathrm{IC}|$ for IC-weighted aggregation). The IC weight ensures that components with higher predictive power contribute more.

The inner product $\langle \mathbf{x}, \mathbf{k}\rangle$ requires $O(W)$ operations per kernel per bar. For $|\mathcal{K}| \leq 20$ and $W = 20$, online scoring costs 400 multiply-accumulate operations per candle—well within real-time constraints.

10 Validation Framework

10.1 Walk-Forward Validation

Algorithm: Walk-Forward Kernel Discovery and Validation

1. Split dataset at $T_{\text{train}} = \lfloor 0.60 \cdot T \rfloor$.
2. **Discovery** on $t \leq T_{\text{train}}$:
For each (e, c): detect events, build $\mathbf{X}^{(e,c)}$, SVD, IC gate.
3. Apply BH-FDR across all in-sample tests.
4. **Validation** on $t > T_{\text{train}}$:
For each OOS event: score with $\mathcal{K}$; record $S^{(e)}$ and $r^{(H)}$.
5. Compute OOS IC; report IC decay ratio.

A kernel is considered robust if

$$\rho_{\text{decay}} := \frac{\mathrm{IC}_k^{\text{OOS}}}{\mathrm{IC}_k^{\text{IS}}} \geq 0.5. \tag{25}$$

10.2 Rolling SVD Stability

Definition 10.1 (Rolling Stability)**.** For rolling windows $\mathcal{D}_1, \ldots, \mathcal{D}_M$ with overlap, let $\mathbf{v}_1^{(m)}$ be the first right singular vector on window m. The stability between consecutive windows is

$$\rho_m := \left|\langle \mathbf{v}_1^{(m)}, \mathbf{v}_1^{(m+1)}\rangle\right|. \tag{26}$$

The overall stability is $\bar{\rho} = (M-1)^{-1}\sum_{m=1}^{M-1}\rho_m$. We require $\bar{\rho} \geq 0.70$.

11 Analytical Basis Dictionary

As a diagnostic, we define an analytical basis dictionary against which data-driven shapes can be compared. Let $\bar{\tau} = \tau/(W-1) \in [0, 1]$.

Polynomial family.

$$\phi_{q1}(\bar{\tau}) = \bar{\tau}^2, \quad \phi_{q2}(\bar{\tau}) = (\bar{\tau} - \tfrac{1}{2})^2, \quad \phi_{q3}(\bar{\tau}) = (1-\bar{\tau})^2. \tag{27}$$

Cubic family.

$$\phi_{c1}(\bar{\tau}) = \bar{\tau}^3, \quad \phi_{c2}(\bar{\tau}) = (\bar{\tau} - \tfrac{1}{2})^3, \quad \phi_{c3}(\bar{\tau}) = (1-\bar{\tau})^3, \quad \phi_{s1}(\bar{\tau}) = 3\bar{\tau}^2 - 2\bar{\tau}^3. \tag{28}$$

The smoothstep ϕ_{s1} (Hermite easing function) satisfies $\phi_{s1}(0) = 0$, $\phi_{s1}(1) = 1$, $\phi'_{s1}(0) = \phi'_{s1}(1) = 0$.

Exponential family $(a \in [3, 8])$.

$$\phi_{e1}(\bar{\tau}) = e^{-a\bar{\tau}}, \quad \phi_{e2}(\bar{\tau}) = e^{-a(1-\bar{\tau})}, \quad \phi_{b1}(\bar{\tau}) = e^{-a(\bar{\tau}-1/2)^2}. \tag{29}$$

Alignment diagnostic. For each SVD-discovered vector $\mathbf{v}_k$ and analytical basis ϕ_i (discretised to W points and L^2-normalised):

$$\rho_{k,i} := \left|\langle \mathbf{v}_k, \hat{\boldsymbol{\phi}}_i \rangle\right|. \tag{30}$$

Values $\rho > 0.85$ indicate strong alignment (interpretable shape); $\rho < 0.5$ indicates a genuinely novel data-driven shape.

12 Empirical Results

We apply the pipeline to BTC perpetual futures on Hyperliquid, a decentralised derivatives exchange, using 100 ms tick data aggregated to 60-second candles.

12.1 Data

Property	**Value**
Asset	BTC-USD perpetual
Exchange	Hyperliquid
Tick frequency	100 ms
Total ticks	7,235,623
Candle frequency	60 s
Total candles	12,059
Parameters	$W = 20$, $N = 20$, $K = 3$, $\alpha = 1.0$, $N_{\text{atr}} = 14$

12.2 Event Detection

Event Type	**Count**	**Rate (per 1000 candles)**
Breakout bull	418	34.7
Breakout bear	522	43.3
Turtle bull	286	23.7
Turtle bear	241	20.0
Trap bull	136	11.3
Trap bear	172	14.3
Total	**1,775**	**147.2**

Breakout events are most frequent (bear > bull asymmetry). Trap events are rarest, consistent with their stricter multi-bar confirmation criteria. All event types exceed the $n_{\min} = 100$ threshold required for stable SVD.

12.3 SVD Stereotypy

Table 1 reports the stereotypy index $\text{SI} = \sigma_1/\sigma_2$ across all (event, channel) pairs.

12.4 Surviving Kernels

From 158 candidates passing $|\text{IC}| > 0.03$, BH-FDR correction at $q = 0.05$ retains **6 kernels** (Table 2).

Three findings merit emphasis:

Event Type	body	upper wick	lower wick	volume
Breakout bull	1.24	1.54	1.80	**2.42**
Breakout bear	1.39	1.55	**1.93**	1.54
Turtle bull	1.02	1.28	**1.59**	1.49
Turtle bear	1.16	1.32	1.36	**1.52**
Trap bull	**1.69**	1.55	1.54	**2.74**
Trap bear	1.17	1.34	1.59	**1.67**

Table 1: Stereotypy index by event type and channel. Bold indicates highest per row. Volume consistently shows the highest stereotypy for breakout and trap events, indicating a more consistent volume signature than price shape. Trap bull volume (SI = 2.74) is the highest across all pairs.

#	Event	Channel	k	IC	p-value	EVR
1	Turtle bull	body	0	$+0.203$	5.4×10^{-4}	12.2%
2	Turtle bull	body	6	-0.201	6.3×10^{-4}	5.6%
3	Turtle bear	lower wick	0	-0.262	3.7×10^{-5}	11.2%
4	Trap bull	upper wick	6	$+0.357$	2.0×10^{-5}	4.7%
5	Trap bull	volume	0	-0.270	1.5×10^{-3}	26.5%
6	Trap bear	body	7	-0.272	3.0×10^{-4}	4.9%

Table 2: Surviving kernels after BH-FDR correction. No breakout kernels survive. Turtle and trap events dominate. The trap bull volume kernel ($k = 0$, EVR = 26.5%) captures a quarter of all trap bull volume variance in a single shape, confirming that volume carries independent predictive information.

1. **No breakout kernels survive.** Despite high EVR (breakout shapes exist and are consistent), they carry no predictive power for forward returns after FDR correction. The market may have already incorporated breakout information by the time the pattern completes.

2. **Multi-channel contribution.** The 6 survivors span three channels (body, lower wick, upper wick, volume), confirming that the multi-channel decomposition captures information absent from the body channel alone.

3. **Moderate IC magnitudes.** All ICs are in the range $[0.20, 0.36]$, appropriate for 10-minute forward returns. The signals are genuine but not strong enough to trade in isolation.

12.5 Alignment with Analytical Basis

All 6 surviving kernels have maximum cosine similarity < 0.5 against all 10 analytical basis functions. The data-driven shapes are genuinely novel—they cannot be approximated by polynomial, cubic, or exponential templates. This validates the SVD-based discovery approach over hand-designed templates.

12.6 Walk-Forward Validation

Train: 7,235 candles (60%); test: 4,824 candles (40%).

Event	Channel	k	IS IC	OOS IC	Decay	Robust
Breakout bull	volume	0	+0.243	−0.010	−0.04	No
Breakout bear	body	10	+0.271	+0.038	0.14	No
Turtle bull	body	7	+0.285	+0.108	0.38	No
Turtle bear	lower wick	0	−0.420	−0.108	0.26	No
Trap bull	upper wick	4	−0.474	+0.126	−0.27	No
Trap bull	volume	0	−0.449	+0.145	−0.32	No
Trap bear	body	4	+0.418	−0.036	−0.09	No
Trap bear	body	5	+0.415	+0.010	0.03	No

No kernels meet the robustness threshold ($\rho_{\text{decay}} \geq 0.50$). However, this is primarily a data volume limitation: with 12,059 total candles split 60/40, trap events (136–172 occurrences) yield only ~50–70 test-set events—insufficient for statistical significance. Turtle-type kernels show partial generalisation (decay ratios 0.26–0.38 with consistent sign), suggesting that with 3–6× more data, OOS robustness may emerge.

12.7 Rolling Stability

Event Type	Mean $\bar{\rho}$	Min $\bar{\rho}$	Status
Trap bull	**0.941**	0.865	Stable
Turtle bull	**0.920**	0.825	Stable
Breakout bear	**0.855**	0.726	Stable
Turtle bear	0.719	0.352	Borderline
Breakout bull	0.692	0.165	Unstable
Trap bear	0.669	0.465	Unstable

The event types with the strongest IC (trap bull, turtle bull) also exhibit the highest temporal stability ($\bar{\rho} > 0.90$). This correlation between predictive power and shape persistence is expected: if a shape is not stable across time, its predictive association with returns cannot be stable either.

13 Discussion

13.1 Methodological Implications

The central finding is that SVD applied to event-aligned candlestick windows discovers statistically significant pattern shapes, but only for reversal-type events (turtle soup and traps), not for momentum-type events (breakouts). This has a natural economic interpretation: breakout patterns are widely traded and rapidly arbitraged, so the shape of a breakout carries no residual predictive information by the time it completes. Turtle-soup and trap patterns, by contrast, involve deceptive price action (false breakouts followed by reversals) that may be harder for the market to price efficiently.

13.2 Multi-Channel Decomposition

The volume channel contributes the single strongest kernel (trap bull volume, EVR = 26.5%, IC = −0.270). This validates the multi-channel approach: had we used only the body channel—as most pattern recognition studies do—this signal would have been missed entirely. The volume signature of a trap event is more stereotyped (SI = 2.74) than its price signature (SI = 1.69), suggesting that volume dynamics carry structural information about the nature of the breakout that is absent from price alone.

13.3 Anti-Overfitting Architecture

The pipeline provides three independent layers of protection:

1. SVD is return-blind: shapes are discovered without access to forward returns.
2. BH-FDR controls the expected fraction of false discoveries at $q = 5\%$ across all tested (e, c, k, H) combinations.
3. Walk-forward OOS validation tests generalisability to unseen data.

13.4 Limitations

1. **Data volume.** With 12,059 candles, rare events (traps: $\sim$150 occurrences) have insufficient test-set counts for walk-forward validation. We estimate 50,000+ candles (3–6 months at 60 s resolution) are needed for robust OOS results.
2. **Single asset.** Results are for BTC perpetual only. Cross-asset universality (ETH, SOL) is untested.
3. **Single exchange.** Hyperliquid is a decentralised exchange with specific microstructure properties. Generalisation to centralised venues remains to be demonstrated.
4. **Stationarity.** The rolling stability analysis shows temporal persistence over the available data, but cannot guarantee persistence across major regime changes.

14 Conclusion

We have presented a complete pipeline for data-driven pattern kernel discovery in financial markets, combining analytical event definitions with SVD-based shape decomposition and IC-gated statistical filtering. The methodology bridges classical technical analysis and modern statistical learning: events are defined by practitioner-interpretable price conditions, but the shapes are discovered by the data rather than prescribed by tradition.

Applied to BTC perpetual futures, the pipeline discovers 6 statistically significant kernels from 158 candidates. The surviving kernels are concentrated in turtle-soup and trap event types, with contributions from multiple candle channels (body, wick, volume). Rolling stability analysis confirms that the strongest kernels persist across different time windows ($\bar{\rho} > 0.90$). The discovered shapes have low alignment with hand-designed analytical basis functions ($\max \rho < 0.5$), demonstrating that the data-driven approach captures structures that an analyst would not have designed a priori.

The primary limitation is data volume: walk-forward OOS validation does not yet confirm robustness at the $\rho_{\text{decay}} \geq 0.50$ threshold. This is a data limitation rather than a methodological failure—in-sample ICs are strong ($|\text{IC}|$ up to 0.47) and turtle-type kernels show partial OOS generalisation. Future work will extend the analysis to multiple assets and longer time horizons.

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